

FIFA 19 Player Analysis and Machine Learning Report

1. Dataset Description

The analysis utilizes the **FIFA 19 Player Dataset** (fifa_eda.csv). This dataset contains comprehensive attributes for football players featured in the video game FIFA 19.

- **Initial Dimensions:** 18,207 rows and 18 columns.
 - **Key Attributes:**
 - **Identity:** ID, Name, Nationality, Club.
 - **Performance Metrics:** Overall Rating, Potential.
 - **Financials:** Value, Wage, Release Clause.
 - **Physical/Technical:** Age, Height, Weight, Preferred Foot, Skill Moves, International Reputation.
 - **Data Cleaning & Preprocessing:**
 - Dropped non-essential columns for modeling: ID, Contract Valid Until, and Club.
 - Handled missing values: Filled missing entries in the Value column with the mean.
 - **Target Variable Creation:** A new binary target variable HighOverall was created. Players with an Overall rating of **80 or higher** were classified as **1 (High Overall)**, while those below 80 were classified as **0 (Others)**.
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2. Exploratory Data Analysis (EDA) Summary

The exploratory phase focused on understanding the distribution of player ratings and relationships between physical/monetary attributes and player quality.

- **Rating Distribution:** The "Overall" rating follows a near-normal distribution, centered around a rating of 66.
- **Correlations:** There is a very strong positive correlation between a player's "Potential" and their current "Overall" rating.
- **Categorical Analysis:** The player's preferred foot (Left vs. Right) does not show a statistically significant impact on the median Overall rating.

3. Visualizations

Distribution of Player Ratings

Description: The histogram above illustrates the frequency distribution of the Overall rating. The data is roughly bell-shaped (normal distribution), with the majority of players falling between ratings of 60 and 70. Very few players achieve elite status (ratings > 85).

Potential vs. Overall Rating

Description: This scatter plot reveals a strong linear relationship between a player's Potential and Overall rating. While there is a tight correlation, the spread increases slightly at higher ratings, indicating that while high potential is necessary for a high overall rating, not all high-potential players have fully realized their rating yet.

Impact of Preferred Foot on Performance

Description: The boxplot compares the distribution of Overall ratings based on the player's Preferred Foot (Left vs. Right). The median ratings (the central line in the box) are nearly identical, suggesting that being left or right-footed does not inherently determine a player's overall quality level in the game.

4. Model Implementation Steps

Feature Selection

The following features were selected to predict the binary target HighOverall:

1. **Age:** Player maturity often correlates with peak performance.
2. **Potential:** The ceiling of a player's ability.
3. **Value & Wage:** Financial valuation is a strong proxy for player skill.
4. **Height & Weight:** Physical attributes.

Preprocessing & Split

- **Target Variable:** The dataset is highly imbalanced.
 - Class 0 (Low/Avg Overall): ~17,652 players.
 - Class 1 (High Overall): ~555 players.

- **Train-Test Split:** The data was split into 80% Training and 20% Testing sets. Crucially, a **stratified split** was used to ensure the proportion of High Overall players remained consistent in both the training and testing sets.

Algorithm

- **Model:** Logistic Regression.
 - **Parameters:** max_iter=500 (to ensure convergence) and multi_class='auto'.
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5. Evaluation Metrics

The model was evaluated using Accuracy, a Confusion Matrix, and a Classification Report.

Confusion Matrix

Description: The heatmap visualizes the model's predictions versus actual values on the test set.

- **True Negatives (3527):** Correctly identified "Other" players.
- **True Positives (98):** Correctly identified "High Overall" players.
- **False Positives (4):** "Other" players incorrectly labeled as Elite.
- **False Negatives (13):** "High Overall" players missed by the model.

Metric Summary

- **Overall Accuracy: 99.53%**
- **Classification Report:**
 - **Majority Class (Others):** Precision 1.00, Recall 1.00.
 - **Minority Class (High Overall):** Precision 0.96, Recall 0.88.

(Note: In the provided notebook code, the labels in the classification report print statement appear to be swapped based on the support count. The interpretation above corrects for this based on the confusion matrix data).

6. Interpretation of Results

The Logistic Regression model performed exceptionally well, achieving near-perfect accuracy.

1. **High Predictive Power:** The features Value and Wage likely contribute significantly to this high accuracy. In FIFA data, a player's market value and wage are algorithmically tied to their overall rating. High-rated players almost exclusively have high wages and values, making the separation between classes linear and distinct.
2. **Handling Imbalance:** Despite the dataset being heavily imbalanced (very few elite players), the model achieved a high F1-score for the minority class. It successfully identified 98 out of 111 elite players in the test set while only misclassifying 4 non-elite players as elite.
3. **Conclusion:** The model is reliable for classifying elite players using financial and potential metrics.